

this, you can install the Flask package by using the following command:

```
$ pip install flask
```

To verify the installation of the Python and Flask version, run the following command, as shown in Figure 1.

```
$ python --version
$ python -m flask --version
```

Creating a sample Flask web application

Now, we are ready to create a sample Flask web application — let's say 'aditya.py'. This will print a sample message 'Welcome to OSFY 2022 Flask App'. The sample code of the created file is shown in Figure 2.

```
aditya.py
1 from flask import Flask
2 application = Flask(__name__)
3 @application.route('/')
4 def OSFY():
5     return 'Welcome to OSFY 2022 Flask App'
6 if __name__ == "__main__":
```

```
7     application.run()
```

A brief explanation of this Flask application source code is given below.

Line 1: In the first line, a Flask class is imported, which is used to create a web application prototype.

Line 2: This line is used to create an instance of the Flask class.

Line 3: `route()` decorator is used to map the OSFY function for the / route. When the function call `OSFY()` is triggered, this will return the user message.

Finally, you can execute this source code by running the command at the terminal.

```
$ python3 filename.py
```

After successful execution of the command, the output can be viewed at the URL `http://127.0.0.1:5000`. Figure 3 shows the Python Flask web output for our created web application.

Launching AWS EC2 server using Elastic Beanstalk

Once a Flask web application is created, we can deploy a web server by launching an AWS EC2

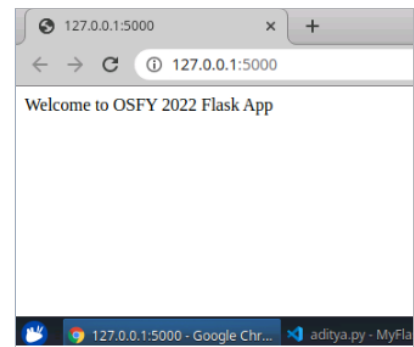


Figure 3: Python Flask web application output

cloud instance using AWS Elastic Beanstalk services. AWS Beanstalk makes the hosting of user web applications on the web servers managed by the AWS (Amazon Web Services) cloud computing platform. Key features like virtual machines, load balancer, etc. are provided by Elastic Beanstalk.

To start with, you need to log into your AWS account and select Elastic Beanstalk. Now create an application, let's say 'OSFYFlaskEC2App', as shown in Figure 4. Once you have created this Flask application, you can see that your web app is running on the AWS Elastic Beanstalk server.

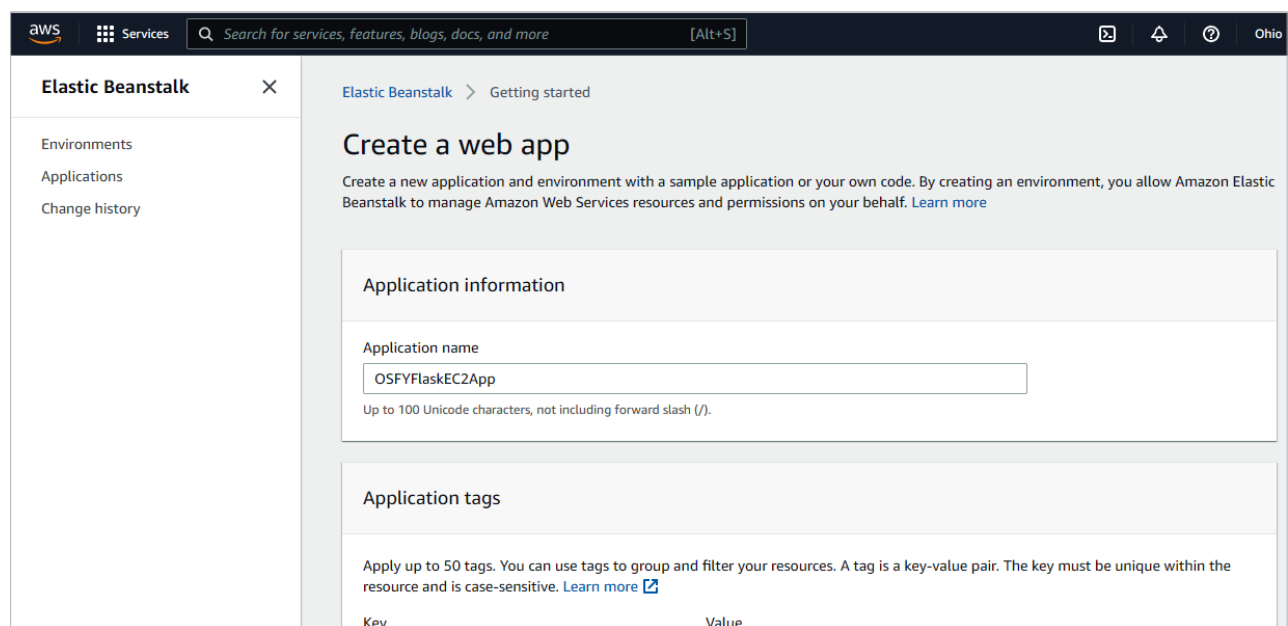


Figure 4: Creating a new application at Elastic Beanstalk

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